YEAR 8 MATHEMATICS

Homeschool Student Workbook

Student: _____

Start date: _____

Parent / tutor: _____

Six units | 12-week introductory program | 144 practice questions | 6 revision tests | Answer key

Designed for Australian Year 8 learners. This is a foundational program, not a complete Year 8 curriculum.

How to use this workbook

Work through one unit every two weeks. Each unit contains three short teaching explanations, a 24-question practice worksheet and a 10-question revision test. Aim for 45-60 minutes of mathematics on four or five days each week.

Suggested two-week routine: Days 1-2 review examples; Days 3-6 complete worksheet in groups of six questions; Days 7-8 correct errors and practise weak areas; Day 9 complete the revision test without help; Day 10 review and plan next steps.

Materials: pencil, ruler, eraser, graph paper, calculator (only when instructed), tape measure and coloured pencils.

Assessment: Aim for at least 8/10 on each revision test. Review errors before moving forward. Keep a separate exercise book for calculations if more room is needed.

12-week progress record

Unit	Weeks	Worksheet /24	Test /10	Completed
1	1-2	_____	_____	_____
2	3-4	_____	_____	_____
3	5-6	_____	_____	_____
4	7-8	_____	_____	_____
5	9-10	_____	_____	_____
6	11-12	_____	_____	_____

UNIT 1: Number Skills, Fractions and Percentages

Learning goal: Work confidently with signed numbers, fractions, decimals, percentage discounts and increases.

Success check: I can explain the methods, show my working and check my answers.

Teacher-led worked examples

Signed numbers. When subtracting a negative, add its positive: 8 - (-5) = 8 + 5 = 13.

Fractions. Find a common denominator: 2/3 + 1/4 = 8/12 + 3/12 = 11/12.

Percentages. 25% of $80 = 0.25 x 80 = $20; discounted price = $80 - $20 = $60.

Try together

Choose one similar example of your own and explain each step aloud.

Independent worksheet - 24 questions

Show working. Do not use the answer key until you finish.

1. Calculate -7 + 12.

2. Calculate -8 - 6.

3. Calculate 9 - (-4).

4. Calculate (-3) x 7.

5. Calculate 24 / (-6).

6. Simplify 12/18.

7. Calculate 3/4 + 1/8.

8. Calculate 5/6 - 1/3.

9. Calculate 2/5 x 3/4.

10. Calculate 3/4 / 1/2.

11. Convert 0.375 to a percentage.

12. Convert 35% to a simplified fraction.

13. Calculate 2.75 x 4.

14. Calculate 9.6 / 0.8.

15. Find 15% of $240.

16. A $120 jacket is reduced by 30%. Find its sale price.

17. Increase $150 by 12%.

18. Find 40% of 65.

19. Write 7/8 as a decimal.

20. Order from smallest to largest: -2, 3, -5, 0.

21. A $50 purchase attracts 10% GST. Find the total.

22. A class has 12 girls and 18 boys. What percentage are girls?

23. A tank is 3/5 full. What percentage is full?

24. You save $18 of $120. What percentage did you save?

UNIT 1: Revision test

Name: _____ Date: _____ Score: _____ / 10

Complete independently. Show working for every calculation.

1. Calculate -7 + 12.

2. Calculate 9 - (-4).

3. Calculate 24 / (-6).

4. Calculate 3/4 + 1/8.

5. Calculate 2/5 x 3/4.

6. Convert 0.375 to a percentage.

7. Calculate 2.75 x 4.

8. Find 15% of $240.

9. Increase $150 by 12%.

10. Write 7/8 as a decimal.

Reflection: Which two skills should I practise next?

UNIT 2: Algebra and Equations

Learning goal: Use variables, simplify expressions, substitute values and solve one- and two-step equations.

Success check: I can explain the methods, show my working and check my answers.

Teacher-led worked examples

Variables. If x = 4, then 3x + 7 = 3 x 4 + 7 = 19.

Collect like terms. 5x + 3x - 2 = 8x - 2.

Solve an equation. 2x + 6 = 20; subtract 6: 2x = 14; divide by 2: x = 7.

Try together

Choose one similar example of your own and explain each step aloud.

Independent worksheet - 24 questions

Show working. Do not use the answer key until you finish.

1. Simplify 5x + 3x - 2.

2. Simplify 7a - 2a + 4.

3. Evaluate 3a + 4 when a = 5.

4. Evaluate 2x squared when x = 3.

5. Solve x + 9 = 21.

6. Solve x - 7 = 15.

7. Solve 4x = 36.

8. Solve x/5 = 8.

9. Solve 2x + 5 = 17.

10. Solve 3x - 4 = 20.

11. Solve $5x + 2 = 27$.

12. Expand $3(x + 4)$.

13. Expand $2(3x - 5)$.

14. Simplify $2x + 5 + 3x - 1$.

15. Solve $7 + x = 2x - 4$.

16. Solve $4(x + 2) = 28$.

17. If $y = 2x + 3$, find y when $x = 6$.

18. If $a = -2$, evaluate a squared $+ 3a$.

19. Write an expression for 5 more than twice n.

20. Write an equation: three times n plus 7 is 31.

21. Solve the equation in Question 20.

22. Three identical notebooks and a $5 pen cost $20. Find one notebook.

23. Solve $6x - 7 = 2x + 9$.

24. Factorise $6x + 12$.

UNIT 2: Revision test

Name: _____ Date: _____ Score: _____ / 10

Complete independently. Show working for every calculation.

1. Simplify 5x + 3x - 2.

2. Evaluate 3a + 4 when a = 5.

3. Solve x + 9 = 21.

4. Solve 4x = 36.

5. Solve 2x + 5 = 17.

6. Solve 5x + 2 = 27.

7. Expand 2(3x - 5).

8. Solve 7 + x = 2x - 4.

9. If y = 2x + 3, find y when x = 6.

10. Write an expression for 5 more than twice n.

Reflection: Which two skills should I practise next?

UNIT 3: Measurement, Area and Volume

Learning goal: Convert units and calculate perimeter, area, circumference and volume.

Success check: I can explain the methods, show my working and check my answers.

Teacher-led worked examples

Rectangle. Area = length x width. For 8 cm by 5 cm: area = 40 cm squared; perimeter = 2(8 + 5) = 26 cm.

Triangle. Area = 1/2 x base x height. For base 12 cm and height 7 cm: 42 cm squared.

Rectangular prism. Volume = length x width x height. For 5 cm x 4 cm x 3 cm: 60 cm cubed.

Try together

Choose one similar example of your own and explain each step aloud.

Independent worksheet - 24 questions

Show working. Do not use the answer key until you finish.

1. Convert 2.5 m to cm.

2. Convert 1800 mL to L.

3. Convert 3.4 km to m.

4. Find perimeter of an 8 cm by 5 cm rectangle.

5. Find area of an 8 cm by 5 cm rectangle.

6. Find area of a 12 cm base, 7 cm height triangle.

7. Find area of a square of side 9 cm.

8. Find perimeter of a square of side 11 cm.

9. Find volume of a 5 x 4 x 3 cm box.

10. Find volume of a 2 m x 3 m x 4 m room.

11. Find circumference of radius 7 cm circle, pi = 3.14.

12. Find area of radius 5 cm circle, pi = 3.14.

13. Convert 0.6 m squared to cm squared.

14. Convert 2500 cm cubed to L.

15. A floor is 4 m x 3 m. How many 1 m squared tiles?

16. Flooring costs $25 per m squared. Cost for 12 m squared?

17. Find area of parallelogram base 10 cm, height 6 cm.

18. Find area of trapezium with parallel sides 6 cm and 10 cm, height 4 cm.

19. Find volume of cube side 4 cm.

20. Find total distance around triangle with sides 7, 8 and 9 cm.

21. A 2 m x 1.5 m wall needs paint at 1 L per 6 m squared. How many L?

22. Convert 75 mm to cm.

23. A 15 m fence has a 2 m gate. How much fencing is needed?

24. A tank holds 2.4 kL. How many litres?

UNIT 3: Revision test

Name: _____ Date: _____ Score: _____ / 10

Complete independently. Show working for every calculation.

1. Convert 2.5 m to cm.

2. Convert 3.4 km to m.

3. Find area of an 8 cm by 5 cm rectangle.

4. Find area of a square of side 9 cm.

5. Find volume of a 5 x 4 x 3 cm box.

6. Find circumference of radius 7 cm circle, pi = 3.14.

7. Convert 0.6 m squared to cm squared.

8. A floor is 4 m x 3 m. How many 1 m squared tiles?

9. Find area of parallelogram base 10 cm, height 6 cm.

10. Find volume of cube side 4 cm.

Reflection: Which two skills should I practise next?

UNIT 4: Geometry and Pythagoras

Learning goal: Use angle rules, coordinate reflections and right-angled triangle calculations.

Success check: I can explain the methods, show my working and check my answers.

Teacher-led worked examples

Angles. Angles on a straight line sum to 180 degrees. A 115 degree angle has a supplement of 65 degrees.

Triangle. Interior angles total 180 degrees. If two angles are 50 and 65 degrees, the third is 65 degrees.

Pythagoras. For a right triangle, a squared + b squared = c squared. With legs 3 m and 4 m, c = square root of 25 = 5 m.

Try together

Choose one similar example of your own and explain each step aloud.

Independent worksheet - 24 questions

Show working. Do not use the answer key until you finish.

1. Find third angle if two triangle angles are 50 and 65 degrees.

2. Find supplement of 115 degrees.

3. Find complement of 38 degrees.

4. Find third angle if angles are 90 and 35 degrees.

5. What is the angle sum of a quadrilateral?

6. What is each angle of a square?

7. Find hypotenuse of right triangle with legs 6 cm and 8 cm.

8. Find hypotenuse with legs 5 cm and 12 cm.

9. Find missing leg if hypotenuse 13 cm and other leg 5 cm.

10. Find hypotenuse of legs 9 m and 12 m.

11. Reflect point (3, -2) in y-axis.

12. Reflect point (4, 5) in x-axis.

13. Translate (2, 3) by 4 right and 2 down.

14. Find sum of angles around a point.

15. Find missing angle around a point: 90, 120, 80, ?.

16. Find exterior angle of regular hexagon.

17. How many lines of symmetry does a rectangle (not a square) have?

18. Find interior angle sum of a pentagon.

19. Find one interior angle of a regular pentagon.

20. Classify a triangle with sides 5, 5, 8.

21. Classify a triangle with angles 90, 40, 50.

22. A ladder reaches 4 m high with its base 3 m from wall. Find ladder length.

23. Find missing leg if hypotenuse 10 m and leg 8 m.

24. Find distance between (0,0) and (3,4).

UNIT 4: Revision test

Name: _____ Date: _____ Score: _____ / 10

Complete independently. Show working for every calculation.

1. Find third angle if two triangle angles are 50 and 65 degrees.

2. Find complement of 38 degrees.

3. What is the angle sum of a quadrilateral?

4. Find hypotenuse of right triangle with legs 6 cm and 8 cm.

5. Find missing leg if hypotenuse 13 cm and other leg 5 cm.

6. Reflect point (3, -2) in y-axis.

7. Translate (2, 3) by 4 right and 2 down.

8. Find missing angle around a point: 90, 120, 80, ?.

9. How many lines of symmetry does a rectangle (not a square) have?

10. Find one interior angle of a regular pentagon.

Reflection: Which two skills should I practise next?

UNIT 5: Ratios, Rates and Linear Graphs

Learning goal: Use proportional reasoning, rates, coordinate tables and straight-line rules.

Success check: I can explain the methods, show my working and check my answers.

Teacher-led worked examples

Ratio. 12:18 simplifies by dividing both parts by 6 to 2:3.

Unit rate. 180 km in 3 hours = 180 / 3 = 60 km per hour.

Linear rule. For y = 2x + 1, x = 0,1,2,3 gives y = 1,3,5,7; gradient is 2.

Try together

Choose one similar example of your own and explain each step aloud.

Independent worksheet - 24 questions

Show working. Do not use the answer key until you finish.

1. Simplify 12:18.

2. Simplify 15:25.

3. Divide $60 in ratio 2:3.

4. Divide 40 sweets in ratio 3:5.

5. A car travels 180 km in 3 h. Find speed.

6. A cyclist travels 45 km in 1.5 h. Find speed.

7. Find y if y = 2x + 1 and x = 4.

8. Find y if y = 3x - 2 and x = 5.

9. Next two terms: 3, 7, 11, 15, ...

10. Find nth-term rule for 4, 7, 10, 13, ...

11. Find gradient between (0,2) and (3,8).

12. Find gradient between (1,1) and (3,7).

13. Find y-intercept of $y = 3x - 5$.

14. Find y-intercept of $y = -2x + 4$.

15. At $4/kg, cost of 2.5 kg?

16. A recipe uses 2 cups cordial to 5 cups water. For 4 cups cordial, water?

17. If 6 pens cost $9, what is cost per pen?

18. If 5 m fabric costs $35, what is price per m?

19. For $y = x + 3$, find y at $x = -2$.

20. For $y = 4x$, find x when $y = 28$.

21. Is (2,5) on line $y = 2x + 1$?

22. Is (3,8) on line $y = 2x + 1$?

23. Find gradient of $y = -3x + 6$.

24. A taxi charges $5 plus $2/km. Cost for 8 km?

UNIT 5: Revision test

Name: _____ Date: _____ Score: _____ / 10

Complete independently. Show working for every calculation.

1. Simplify 12:18.

2. Divide $60 in ratio 2:3.

3. A car travels 180 km in 3 h. Find speed.

4. Find y if y = 2x + 1 and x = 4.

5. Next two terms: 3, 7, 11, 15, ...

6. Find gradient between (0,2) and (3,8).

7. Find y-intercept of y = 3x - 5.

8. At $4/kg, cost of 2.5 kg?

9. If 6 pens cost $9, what is cost per pen?

10. For y = x + 3, find y at x = -2.

Reflection: Which two skills should I practise next?

UNIT 6: Statistics and Probability

Learning goal: Summarise data and calculate theoretical and experimental probabilities.

Success check: I can explain the methods, show my working and check my answers.

Teacher-led worked examples

Mean. For 6, 8, 7, 9, 10: total 40; mean = 40 / 5 = 8.

Median and range. Sort 6, 7, 8, 9, 10. Median = 8; range = 10 - 6 = 4.

Probability. With 3 red and 7 blue balls, P(red) = 3/10 = 30%.

Try together

Choose one similar example of your own and explain each step aloud.

Independent worksheet - 24 questions

Show working. Do not use the answer key until you finish.

1. Mean of 4, 6, 8, 10.

2. Median of 3, 9, 5, 11, 7.

3. Mode of 2, 4, 4, 5, 7.

4. Range of 12, 5, 9, 18.

5. Mean of 5, 5, 10, 20.

6. Median of 2, 4, 6, 8.

7. Mode of 1, 1, 2, 2, 3.

8. Range of -3, 4, 7, 10.

9. Probability of rolling a 6 on fair die.

10. Probability of heads on fair coin.

11. Probability of even number on fair die.

12. Probability of number greater than 4 on fair die.

13. Bag has 3 red and 7 blue balls. P(red)?

14. Same bag: P(blue)?

15. Same bag: P(green)?

16. Same bag: P(red or blue)?

17. In 50 fair coin tosses, expected heads?

18. A coin gives 18 heads in 30 tosses. Experimental P(heads)?

19. A class has 8 left-handed students out of 32. Proportion?

20. Probability 0.2 as percentage?

21. Probability 75% as fraction?

22. What is total probability of all outcomes?

23. Scores: 4, 5, 5, 6, 10. Find mean.

24. Scores: 4, 5, 5, 6, 10. Find median.

UNIT 6: Revision test

Name: _____ Date: _____ Score: _____ / 10

Complete independently. Show working for every calculation.

1. Mean of 4, 6, 8, 10.

2. Mode of 2, 4, 4, 5, 7.

3. Mean of 5, 5, 10, 20.

4. Mode of 1, 1, 2, 2, 3.

5. Probability of rolling a 6 on fair die.

6. Probability of even number on fair die.

7. Bag has 3 red and 7 blue balls. P(red)?

8. Same bag: P(green)?

9. In 50 fair coin tosses, expected heads?

10. A class has 8 left-handed students out of 32. Proportion?

Reflection: Which two skills should I practise next?

Parent / tutor answer key

Answers correspond to the 24-question worksheet in each unit. Revision tests use worksheet questions 1, 3, 5, ... 19 in that order.

Unit 1: Number Skills, Fractions and Percentages

1. 5
2. -14
3. 13
4. -21
5. -4
6. 2/3
7. 7/8
8. 1/2
9. 3/10
10. 3/2 or 1 1/2
11. 37.5%
12. 7/20
13. 11
14. 12
15. $36
16. $84
17. $168
18. 26
19. 0.875
20. -5, -2, 0, 3
21. $55
22. 40%
23. 60%
24. 15%

Revision test answers

1. 5 | 2. 13 | 3. -4 | 4. 7/8 | 5. 3/10 | 6. 37.5% | 7. 11 | 8. $36 | 9. $168 | 10. 0.875

Unit 2: Algebra and Equations

1. 8x - 2
2. 5a + 4
3. 19
4. 18
5. x = 12
6. x = 22
7. x = 9
8. x = 40
9. x = 6
10. x = 8
11. x = 5

12. 3x + 12

13. 6x - 10

14. 5x + 4

15. x = 11

16. x = 5

17. 15

18. -2

19. 2n + 5

20. 3n + 7 = 31

21. n = 8

22. $5

23. x = 4

24. 6(x + 2)

Revision test answers

1. 8x - 2 | 2. 19 | 3. x = 12 | 4. x = 9 | 5. x = 6 | 6. x = 5 | 7. 6x - 10 | 8. x = 11 | 9. 15 | 10. 2n + 5

Unit 3: Measurement, Area and Volume

1. 250 cm
2. 1.8 L
3. 3400 m
4. 26 cm
5. 40 cm squared
6. 42 cm squared
7. 81 cm squared
8. 44 cm
9. 60 cm cubed
10. 24 m cubed
11. 43.96 cm
12. 78.5 cm squared
13. 6000 cm squared
14. 2.5 L
15. 12 tiles
16. $300
17. 60 cm squared
18. 32 cm squared
19. 64 cm cubed
20. 24 cm
21. 0.5 L
22. 7.5 cm
23. 13 m
24. 2400 L

Revision test answers

1. 250 cm | 2. 3400 m | 3. 40 cm squared | 4. 81 cm squared | 5. 60 cm cubed | 6. 43.96 cm | 7. 6000 cm squared | 8. 12 tiles | 9. 60 cm squared | 10. 64 cm cubed

Unit 4: Geometry and Pythagoras

1. 65 degrees
2. 65 degrees
3. 52 degrees
4. 55 degrees
5. 360 degrees
6. 90 degrees
7. 10 cm
8. 13 cm
9. 12 cm
10. 15 m
11. (-3, -2)
12. (4, -5)
13. (6, 1)
14. 360 degrees

15. 70 degrees

16. 60 degrees

17. 2

18. 540 degrees

19. 108 degrees

20. Isosceles

21. Right-angled

22. 5 m

23. 6 m

24. 5 units

Revision test answers

1. 65 degrees | 2. 52 degrees | 3. 360 degrees | 4. 10 cm | 5. 12 cm | 6. (-3, -2) | 7. (6, 1) | 8. 70 degrees | 9. 2 | 10. 108 degrees

Unit 5: Ratios, Rates and Linear Graphs

1. 2:3
2. 3:5
3. $24 and $36
4. 15 and 25
5. 60 km/h
6. 30 km/h
7. 9
8. 13
9. 19, 23
10. 3n + 1
11. 2
12. 3
13. -5
14. 4
15. $10
16. 10 cups
17. $1.50
18. $7
19. 1
20. 7
21. Yes
22. No
23. -3
24. $21

Revision test answers

1. 2:3 | 2. $24 and $36 | 3. 60 km/h | 4. 9 | 5. 19, 23 | 6. 2 | 7. -5 | 8. $10 | 9. $1.50 | 10. 1

Unit 6: Statistics and Probability

1. 7
2. 7
3. 4
4. 13
5. 10
6. 5
7. 1 and 2
8. 13
9. 1/6
10. 1/2
11. 1/2
12. 1/3
13. 3/10
14. 7/10
15. 0

16. 1

17. 25

18. 3/5 or 60%

19. 1/4 or 25%

20. 20%

21. 3/4

22. 1 or 100%

23. 6

24. 5

Revision test answers

1. 7 | 2. 4 | 3. 10 | 4. 1 and 2 | 5. 1/6 | 6. 1/2 | 7. 3/10 | 8. 0 | 9. 25 | 10. 1/4 or 25%

Parent teaching notes and next steps

Mark for method, not only the final number. Ask your student to say what each step means. For a wrong answer, first ask: "What was your first step?" and "How could we check this?"

Support: Use smaller numbers, visual fractions, number lines and physical measurements. Provide one fully worked example beside a new question.

Extension: Ask the student to invent a similar problem, solve it two ways, or explain why a method works.

Curriculum coverage note: These six units provide a practical foundation. To cover the broader Australian Curriculum Year 8 mathematics requirements, add explicit study of integers and rational numbers, exponent laws, irrational numbers, algebraic expansion and factorisation, financial maths, geometric reasoning, transformations, data displays and investigations as appropriate.

Weekly notes: